

FULL-STACK AI ENGINEER

A practical roadmap to balance Gym/Business + DSA + GenAI + Agentic AI + System Design

Your fixed principle	Consistency beats intensity
Morning block	5:00 AM – 7:30 AM
Business block	8:00 AM – 7:00 PM
Career target	Full-Stack AI Engineer
Core daily habit	1 hour DSA

Remember: You do not need 5–6 hours of study every day. A focused 2–2.5 hour morning routine maintained for many months is enough to create strong progress.

1. The Big Picture

You are building two things at the same time: a gym/business during the day and a technology career for the long term. The goal is not to study everything every day. The goal is to protect a small, high-quality morning learning block and follow a clear sequence.

Your career direction: Full Stack Developer → AI-enabled Full Stack Developer → Full-Stack AI Engineer.

Your existing foundation: React, JavaScript, Node.js, Express, MongoDB, REST APIs and related full-stack skills.

New layer: DSA + Generative AI + RAG + Agentic AI + System Design.

The target skill stack

React + Node.js + MongoDB + APIs + DSA + LLMs + Embeddings + Vector Databases + RAG + Tool Calling + Agents + System Design + Deployment.

2. Your Daily Routine

Time	Activity	Purpose
5:00–5:10	Wake up, water, freshen up	Start calmly
5:10–6:10	DSA – 1 hour	Interview problem-solving
6:10–7:10	Main technology topic	GenAI / Agentic AI / System Design
7:10–7:30	Revision + notes + plan	Retention
7:30–8:00	Breakfast / preparation	Transition
8:00–7:00	Gym / business	Primary daytime responsibility
7:00–8:00	Dinner + rest	Recovery
8:00–8:30	Optional light revision	Only if energy is good
~9:30–10:00	Sleep	Protect the next morning

Rule: If the evening is exhausting, skip the 8:00–8:30 study session. Do not sacrifice sleep for extra study time.

3. Weekly Learning Schedule

Day	5:10–6:10	6:10–7:10
Monday	DSA	Generative AI
Tuesday	DSA	System Design
Wednesday	DSA	GenAI + coding/building
Thursday	DSA	Agentic AI
Friday	DSA	System Design
Saturday	DSA / revision	Project building
Sunday	DSA / revision	Revision + mock interview

4. DSA – Your Non-Negotiable 1 Hour

Keep DSA daily because interview preparation benefits from repetition. Use the hour actively instead of watching solutions for the whole session.

- **10 min:** Review yesterday's problem.
- **35 min:** Attempt a new problem yourself.
- **10 min:** Compare with the optimal approach and understand why it works.
- **5 min:** Write the pattern, mistake and key idea in your notes.

Suggested progression: Arrays → Strings → Hashing → Two Pointers → Sliding Window → Stack/Queue → Linked List → Binary Search → Trees → Heap → Graphs → Dynamic Programming.

Rule: Never miss twice. Missing one day is normal; returning the next day is the habit.

5. Generative AI Roadmap

Do not jump directly into advanced agent frameworks. First understand the building blocks.

- What is Generative AI and what is an LLM?
- Tokens and context windows
- Prompt engineering
- LLM APIs
- Structured output
- Embeddings
- Vector databases
- RAG (Retrieval-Augmented Generation)
- Function/tool calling
- Evaluation and reliability

Simple mental model: Your application sends useful context + instructions to an LLM, the LLM generates an answer, and your application controls what data and tools are available.

6. RAG – Understand the Flow

RAG allows an AI application to retrieve relevant information before generating an answer.

Step	What happens
1. Documents	Collect PDFs, FAQs, website content or other knowledge.
2. Chunking	Break large documents into smaller useful pieces.
3. Embeddings	Convert text into numerical representations.
4. Vector DB	Store embeddings so similar content can be retrieved.
5. Retrieval	Find relevant chunks for the user's question.
6. LLM	Give the question + retrieved context to the model.
7. Answer	Generate a grounded response.

7. Agentic AI Roadmap

Learn Agentic AI after you understand the GenAI fundamentals. An agent is an application that can use an LLM to decide or coordinate actions using available tools and information.

Mental model: User → Agent → Understand task → Select tool → Execute → Observe result → Decide next step → Final answer.

- Tool/function calling
- Agent workflows
- Planning and multi-step tasks
- Memory
- RAG agents
- Tool selection
- Multi-agent patterns
- Agent evaluation and safety
- Human approval for sensitive actions

Important: Do not measure progress by the number of frameworks you know. Measure it by whether you can build and explain an agentic workflow.

8. System Design Roadmap

System design teaches you how to think about building software that can handle more users, more data and more traffic.

- Requirements: functional and non-functional
- API design
- Database design
- SQL vs NoSQL
- Caching
- Load balancing
- CDN
- Queues and asynchronous processing
- Rate limiting
- Authentication and authorization
- Microservices
- Observability
- Scalability and reliability

Basic architecture mental model:

User → Load Balancer → API Servers → Cache / Database → Message Queue → Workers → External Services

9. Six-Month Roadmap

Period	Main goal	Outcome
Month 1–2	DSA + GenAI fundamentals	Understand LLMs, APIs, prompts, embeddings and vector DB basics.
Month 3	RAG + AI applications	Build a document/PDF Q&A application.
Month 4	Agentic AI	Build an agent with tools, memory or multi-step workflow.
Month 5	System Design	Design scalable full-stack systems and explain trade-offs.
Month 6	Interview mode	DSA + system design + AI projects + mock interviews.

10. Projects That Connect Everything

Use projects to combine your existing full-stack skills with AI instead of learning AI only through tutorials.

Project	What it demonstrates
AI Resume Analyzer	React + Node + LLM + structured output
PDF / Document Q&A	Embeddings + Vector DB + RAG
AI Shopping Assistant	RAG + tool calling + e-commerce APIs
AI Research Assistant	Agent + tools + multi-step workflow

Project	What it demonstrates
AI Support System	Authentication + RAG + agent + database + system design

11. How to Study Each Morning

Use the 3-step learning loop:

- **Learn:** Understand one concept in simple language.
- **Build:** Write code or draw the architecture yourself.
- **Explain:** Explain it aloud as if an interviewer asked you.

If you cannot explain a topic without looking at notes, it is not finished yet.

12. Interview Preparation

For each topic, prepare four things:

- **What:** What is it?
- **Why:** Why do we need it?
- **How:** How does it work internally or in an application?
- **Project:** Where did you use it and what problem did it solve?

For system design, also add: **trade-offs, bottlenecks, scaling strategy and failure handling.**

For AI projects, also add: **data flow, prompt/context strategy, hallucination controls, latency, cost and evaluation.**

13. What Not to Do

- Do not start five courses at the same time.
- Do not study GenAI, Agentic AI and System Design deeply every day.
- Do not skip DSA for weeks and try to catch up on weekends.
- Do not spend all your time watching videos without coding.
- Do not sacrifice sleep to increase study hours.
- Do not learn every new AI framework just because it is trending.
- Do not abandon your existing React/Node full-stack foundation.

14. Your Weekly Scorecard

Habit	Target
DSA	7 × 1 hour
GenAI	2 focused sessions
Agentic AI	1 focused session
System Design	2 focused sessions
Project	1 meaningful build session

Habit	Target
Mock interview / revision	1 session
Sleep	Protect consistent bedtime

15. Your Long-Term Career Picture

Your path does not need to be a complete restart. You can use the full-stack skills you already have and add AI capabilities on top.

Target profile:

React + Node.js + MongoDB + REST APIs + DSA + System Design + LLM APIs + RAG + Vector DB + Tool Calling + Agentic AI + Deployment.

This combination gives you a practical direction toward Full-Stack AI Engineer roles while keeping your existing development skills valuable.

16. The One-Page Rule to Remember

Every morning: 1 hour DSA + 1 hour main tech skill + 20 minutes revision.

Every week: Learn → Code → Build → Explain → Revise.

Every month: Finish one meaningful milestone.

Every 3 months: Have at least one project you can confidently explain in an interview.

Every year: Consistency matters more than temporary intensity.

YOUR DAILY MANTRA

5 AM: Build your career.

8 AM–7 PM: Build your business.

Evening: Recover.

Next morning: Repeat.

Goal: Become a Full-Stack AI Engineer without giving up on the gym/business you are building.